

20. Application-based: A flowchart accepts a set of integers, stores them in a data structure, searches for a specified element, and displays its position. Identify the appropriate data structure and convert the complete flowchart into a C program.

Appropriate Data Structure: Array

The flowchart accepts a set of integers and stores them in an **array**. It then accepts a search element and compares it with each array element using a loop. If the element is found, its position is displayed; otherwise, “**Element not found**” is displayed.

```
#include <stdio.h>
```

```
int main() {
```

```
    int a[100], n, key, i, pos = -1;
```

```
    printf("Enter number of elements: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter elements:\n");
```

```
    for(i = 0; i < n; i++)
```

```
        scanf("%d", &a[i]);
```

```
    printf("Enter element to search: ");
```

```
    scanf("%d", &key);
```

```
    for(i = 0; i < n; i++) {
```

```
        if(a[i] == key) {
```

```
            pos = i + 1;
```

```
            break;
```

```
        }
```

```
    }
```

```
    if(pos != -1)
```

```
        printf("Element found at position %d", pos);
```

```
    else
```

STDIN

10 20 30 40 50

30

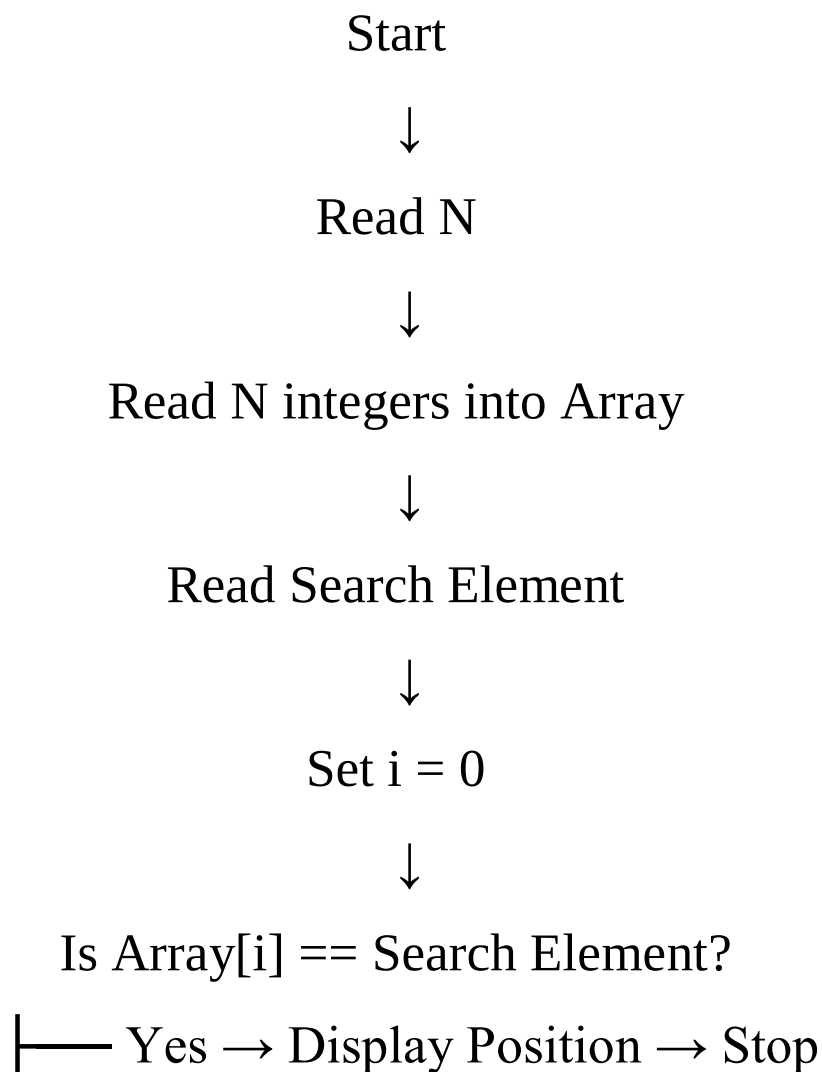
Output:

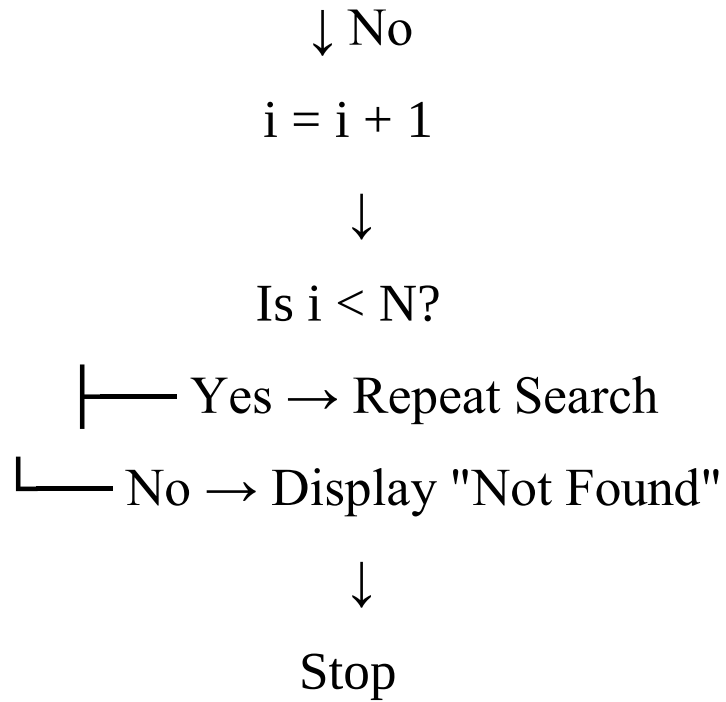
Enter number of elements: Enter elements:

Enter element to search: Element found at position 6

1. Flowchart to Code Conversion – Data Structures

The **array** is the appropriate data structure for storing a set of integers. The flowchart can be converted into C code using an array, loop, and if condition. The program stores the integers, searches for the specified element, and displays its position.





Data Structure Used: Array

Searching Technique: Linear Search

Time Complexity: $O(n)$